INCH-POUND
MIL-PRF-14409/14L
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PERFORMANCE SPECIFICATION SHEET

# CAPACITOR, VARIABLE (PISTON TYPE, TUBULAR TRIMMER), STYLES PC27, PC28, PC29

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification sheet and MIL-PRF-14409.

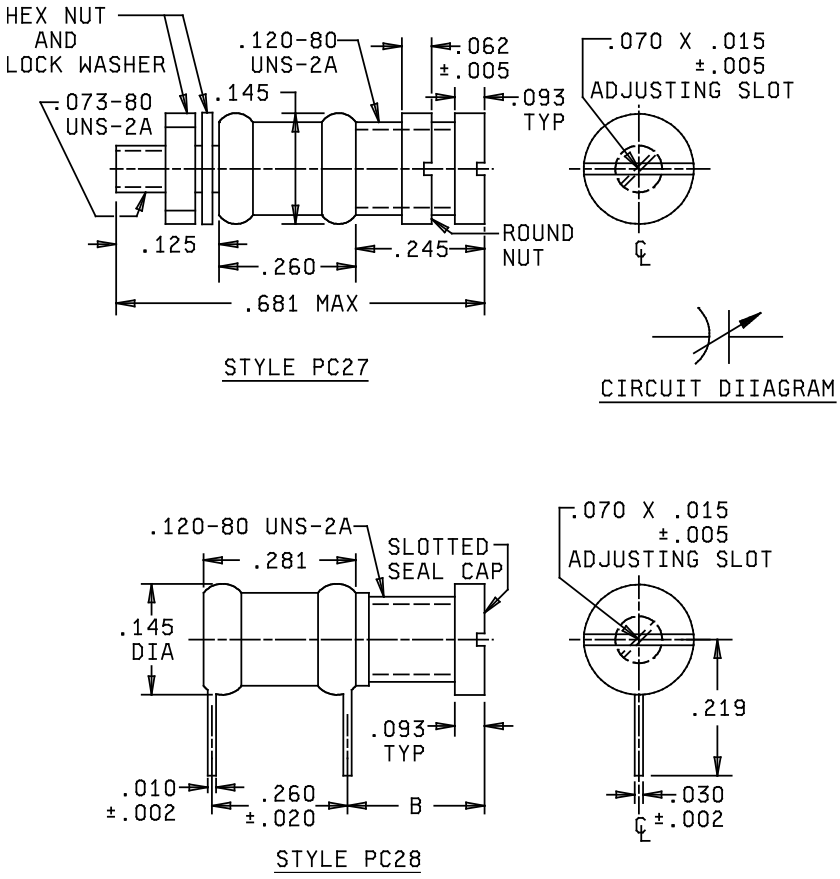
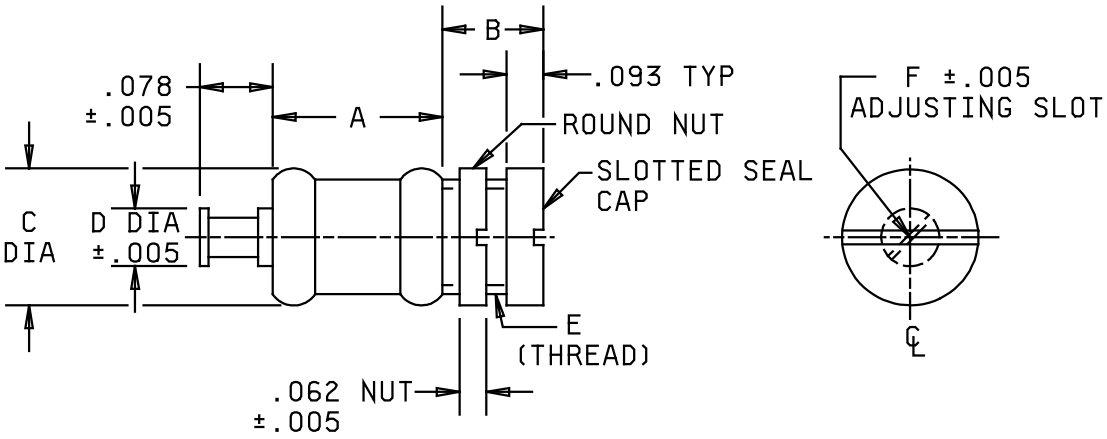



FIGURE 1. Dimensions and configuration.



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STYLE PC29

| Inches | mm | Inches | mm |
|---|---|---|---|
| .002 | 0.05 | .093 | 2.36 |
| .005 | 0.13 | .120 | 3.05 |
| .010 | 0.25 | .125 | 3.18 |
| .015 | 0.38 | .145 | 3.68 |
| .020 | 0.51 | .219 | 5.56 |
| .030 | 0.76 | .245 | 6.22 |
| .062 | 1.57 | .260 | 6.63 |
| .070 | 1.78 | .281 | 7.14 |
| .073 | 1.85 | .681 | 17.30 |
| .078 | 1.98 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .016 inch (0.41 mm).

FIGURE 1. Dimensions and configurations - Continued.



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TABLE I. Capacitor characteristics.

| Part or indentifying number (PIN) | Capacitance range | | Q Min | Dimensions 1/ | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Min (pF) | Max (pF) | | A | B ± .020 (0.51) | C | D | E | F |
| PC27J3R5 | 0.35 | 3.5 | 10,000 | | | | | | |
| PC28J3R5 | 0.35 | 3.5 | 10,000 | | .245 (6.22) | | | | |
| PC28J050 | 0.5 | 5.0 | 7,500 | | .281 (7.14) | | | | |
| PC29J3R5 | 0.35 | 3.5 | 10,000 | .266 (6.76) | .250 (6.35) | .145 (3.68) | .047 (1.19) | .120-80 UNS-2A | .070 x .015 (1.78) (0.38) |
| PC29J050 | 0.5 | 5.0 | 7,500 | .266 (6.76) | .281 (7.14) | .145 (3.68) | .047 (1.19) | .120-80 UNS-2A | .070 x .015 (1.78) (0.38) |
| PC29J100 | 0.8 | 10.0 | 7,500 | .281 (7.14) | .249 (6.32) | .220 (5.59) | .062 (1.57) | .190-64 UNS-2A | .093 Min x .020 (2.36) (0.51) |

1/ Metric equivalents are shown in parentheses for general information only.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

DC voltage rating: 250 volts.

Construction: Alumina ceramic.

Dielectric: Air.

Capacitance range: See table I.

Insulation resistance: Not less than $10^6$ megohms at room ambient temperature and not less than $10^5$ megohms at +125°C.

Test condition - 250 volts ±10 percent.

Quality factor (Q): See table I (measured at frequency of 100 MHz).

Driving torque: Greater than or equal to 0.3 ounce-inches and less than or equal to 4.0 ounce-inches from -55°C through +125°C.

Thermal shock: MIL-STD-202-107, test condition A, except step 3 temperature shall be +125° ±5°C.

Immersion: MIL-STD-202-104, test condition A.

Resistance to soldering heat: Applicable to style PC28 and PC29 only.

Seal: Applicable, with seal cap on.



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Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-14409, this document references the following:

MIL-STD-202-104
MIL-STD-202-107

Custodians:
Army – CR
Navy - EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2021-003)

Review activities:
Navy - AS, MC, OS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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